AMS 2026

Challenge Document

University of Alberta High Speed Rail Group

University of Alberta

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Prepared for Albertaloop Mechatronics Showdown 2026, Challenge Document

Release Authorization [Need for public and condition release docs]

Name	Title	Signature
[Senior Leadership only]	[Title]	
[Senior Leadership only]	[Title]	

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List of Authors

Name	Title
Yuan Liu	Admin Lead and AMS Co-Organizer
Kendall Li	Propulsion Co-lead
[Author 2] (add/subtract rows if needed)	[Title]

List of Peer Reviewers

Name	Title
Jaspreet Chhabra	Co-President
Raymond Vu	Senior Mechanical Team Member
[Peer Revewier 2] (use section if/as needed)	[Title]

1. Rules and Regulations

1.1. Teams are only permitted to use the materials provided per section 4. Use of additional hardware and spare parts outside of the available materials specified in section 4 will be given one warning. Teams caught with additional hardware/parts and have an existing warning will result in the disqualification of the entire team.

1.2. Internet resources such as the Arduino Forum, artificial intelligence tools, and YouTube may be used, provided they do not interfere with or disrupt other participants.

1.3. Use of the ELKO Garage or other shops is not permitted by the AlbertaLoop team is strictly prohibited and will result in an immediate disqualification.

1.4. All participants are prohibited from taking any hardware/material outside of the challenge room (W2-090 / W2-110). This includes all the provided tools and materials listed in section 4.

1.5. In the event of a team challenging the validity of the challenge constraints, it will be up to a counsel of event organizers to determine whether the challenge is valid or not.

2. Point System

2.1. The total amount of points one team can acquire is 100 points

2.2. The minimum amount of points a team may receive per challenge is zero points. Therefore, there will be no deduction from points due to negative points received.

2.3. The maximum number of point attainable per evaluation/challenge is as follows

2.3.1. Balance Beam = 15 Points

2.3.2. Manual Maze Run = 20 Points

2.3.3. Autonomous Maze Run = 30 Points

2.3.4. Design Evaluation = 60 Points

3. Deliverables

3.1. Presentations (Optional)

3.1.1. All teams may develop a slide deck that effectively communicates their design process, rationale, and solutions.

3.1.2. Teams may use any power point software they prefer, but google slides is recommended for the ease of access

3.1.3. Due to time constraints, all presentations will be limited to a maximum of 5 minutes in duration.

3.1.4. Judging criteria will be provided for all teams to reference

3.2. Robot

3.2.1. All teams must design and construct a fully functioning robot with the materials and tools provided

3.2.2. The robot must be able to complete the challenges listed in section 5.

3.2.3. All coding, wiring, and construction will be completed primarily by your team members. AlbertaLoop and Mechatronics members will be available to assist if needed.

4. Materials

The following materials will be provided as per the following table. Keep in mind the items that are listed with "communal" quantities are limited and for all teams to share (other than batteries, fixtures, and jumpers).

Table 1 - Supply and materials list

ARDUINO KIT		BOX ITEMS		COMMUNAL ITEMS
Item	**Quantity**	**Item**	**Quantity**	**Item**
MEGA Controller Board	1	Base Plate	1	Nuts and Bolts and Screws
LCD1602 Module (With Pin header)	1	L-Brackets	10	Additional 3D printed parts
RC422 RFID Module	1	4-battery Mounts	2	Jumpers
Prototype Expansion Module	1	Motor Drivers	2	Batteries: AA, 9V
Mini Bread Board	1	Large Servo MG996R	1	Scissors
Power Supply Module	1	Motors (yellow)	4	Tape: Electrical, Cello, Masking
GY-521 Module	1	Wheels (yellow)	4	Magnetic Plates
Servo Motor SG90	1	Large Servo Mounts (black)	2	Long USB type A to B
Stepper Motor	1	Small Servo Mounts (blue)	2	Limit Switches
ULN Stepper Motor Driver Module	1	Servo Arm Plate	2	Zipties
HC-SR501 PIR Motion Sensor Module	1	Ultrasonic Mount	2	Additional Servos (Big and Small)
Sound Sensor Module	1	2x8 plate	10	Breadboards
Water Level Detection Sensor Module	1	1x8 plate	10	And more….
Ultrasonic Sensor	1	Allen Keys	1 set	
DS1307 RTC Module	1	Adjustable Small Wrench	1	
Rotary Encoder Module	1	Percision Screw Drivers	1 set	
DHT11Temperature and Humidity Module	1	Multimeter	1	
IR Reciever Module	1	Phillips Screwdriver	1	
Remote Control	1	Pliers	1	
MAX7219 Module	1	Double Ended Wrench	1	
1 Digit 7-Segment Display	1	Lan Cable	1	
4-Digit 7-Segment Display	1	Controller	1	

L293D	1	Connection Board	1		
Fan Blade and 3-6 Motor 1PC	1				
Membrane Switch Module 1PC	1				
830 Tie-Points Breadboard 1PC	1				
9V Battery	1				
Snap-on Connector Clip	1				
9V1A Adapter 1PC	1				
USB Cable 1PC	1				
Small component box	1				

5. Challenges

5.1. Balance Beam

The purpose of this challenge is to demonstrate your robots capability in utilizing automation in order to balance on the provided beam. Please note that this event may be done manually as well, but you will receive no points for automation as per the event scoring criteria.

The challenge restraints are listed as follows:

5.1.1. There will be a maximum time limit of two minutes. The timer will commence once the robot begins to move

5.1.2. A standardized wooden balance beam with a traction aid will be used for the challenge. You may inspect the balance beam.

5.1.3. The robot will start partially off the balance beam, with its back wheels off the board and touching the rear edge of the beam.

5.1.4. The board will be considered "balanced" when both sides of the balance beam are suspended in the air for a maximum of five consecutive seconds without touching the ground. The five seconds will be included in your team's total time.

5.1.5. Automation points will be given based on what type of system was utilized for this event up to the judge's discretion. Points will be awarded for all teams following the AMS marking rubric

5.2. Maze Navigation (Manual)

The purpose of this challenge is to evaluate the robot's ability to be manually maneuvered through the designated maze, as shown in *Figure 1*, using only the controller provided by the event organizers.

The challenge restraints are listed as follows:

5.2.1. There will be a maximum time limit of 4 minutes. The timer will commence once the robot begins to move.

5.2.2. All robots will begin in a designated start section illustrated in *Figure 1*

5.2.3. The maze shall be constructed from wooden planks arranged as shown in *Figure 1*. The floor surface shall consist of ECERF carpet.

5.2.4. Checkpoints shown in *Figure 1* can be utilized with the following restrictions:

5.2.4.1. The entire robot shall be fully inside the checkpoint zone in order to be eligible to restart from that checkpoint.

5.2.4.2. When restarting, teams are permitted to restart from the farthest checkpoint reached. The entire robot must sit inside the checkpoint zone behind the line into the next zone. In the event that a robot is not able to fit inside the checkpoint zone, it is permitted that the back portion of the robot can be inside of the zone prior to the checkpoint, sitting behind the line into the next zone

5.2.4.3. The timer will not stop when teams are restarting their robot at their respective checkpoint.

5.3. Maze Navigation (Automation)

This challenge is intended to encourage teams to investigate and apply automotive approaches to the maze-navigation problem.

The challenge restraints are listed as follows:

5.3.1. There will be a maximum time limit of 6 minutes. The timer will commence once the robot begins to move.

5.3.2. All robots will begin in a designated start section illustrated in *Figure 1*

5.3.3. The maze shall be constructed from wooden planks arranged as shown in *Figure 1*. The floor surface shall consist of ECERF carpet.

5.3.4. Checkpoints shown in *Figure 1* can be utilized with the following restrictions:

5.3.4.1. The entire robot shall be fully inside the checkpoint zone in order to be eligible to restart from that checkpoint.

5.3.4.2. When restarting, teams are permitted to restart from the farthest checkpoint reached. The entire robot must sit inside the checkpoint zone behind the line into the next zone. In the event that a robot is not able to fit inside the checkpoint zone, it is permitted that the back portion of the robot can be inside of the zone prior to the checkpoint, sitting behind the line into the next zone

5.3.4.3. The timer will not stop when teams are restarting their robot at their respective checkpoint.

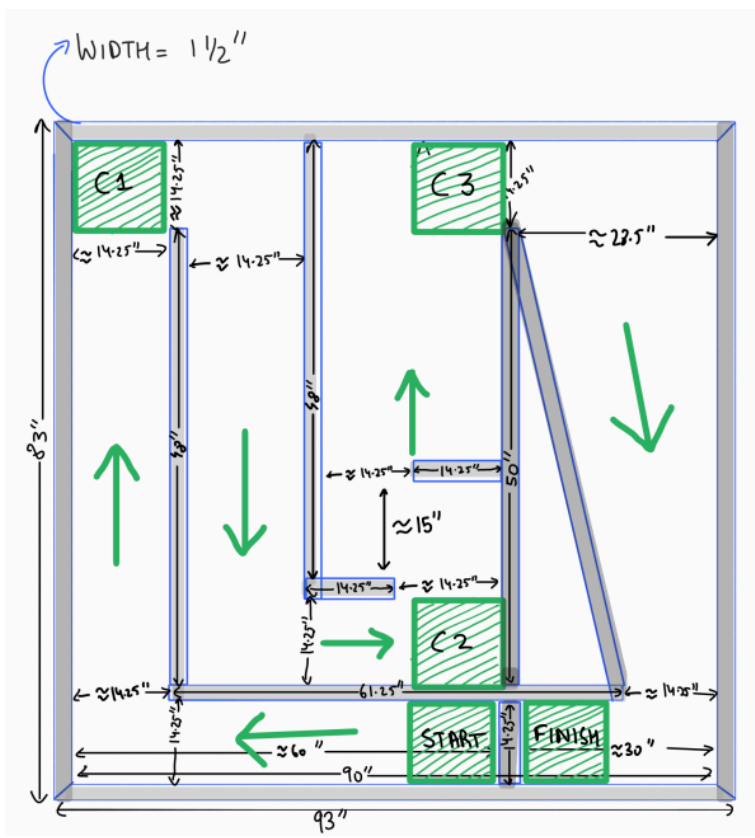

Figure 1. Dimensioning Of The Maze

6. Scoring

The scoring distribution is defined in the AlbertaLoop Mechatronics Showdown 2026 Marking Rubric.